



BITS Pilani

Embedded System Design

Sangmeshwar Kendre



AEZG512/AELZG512/DMZG512
Embedded System Design
Remote Lab Contact Session-02

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- Expt.-2: Implementation of Delay Generation Using Timer (LPTMR) on the S32K144 Microcontroller



Expt.-1: Implementation of Digital Output Control Using GPIO to Blink an LED on the S32K144 Microcontroller board.

- **Objective**

- ✓ Develop an Embedded C application using S32 SDK GPIO / Pins driver to configure a digital output pin (PTD0/PTD15/PTD16) of the S32K144 microcontroller and blink an LED at a visible rate.

- **Problem Description**

- ✓ The S32K144 microcontroller must control onboard LED using a GPIO pin (PTD0/PTD15/PTD16). The selected GPIO pin shall be configured as a digital output. The application shall toggle the LED continuously using a software delay or SDK delay function so that the ON and OFF states are clearly visible.

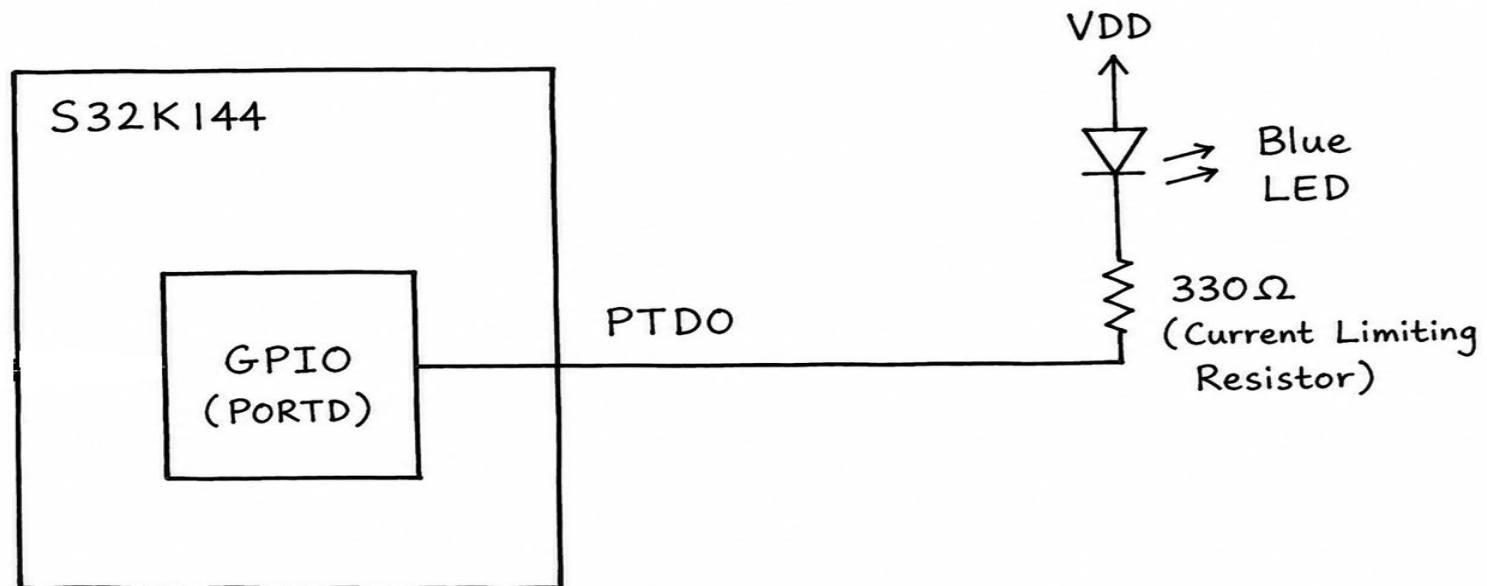
- **Functional Requirements**

- Configure the required system clock using SDK clock configuration.
- Enable and configure the selected LED pin as GPIO output.
- Initialize the pin state to a known logic level.
- Toggle the LED continuously inside an infinite loop.
- Use a delay to make blinking visible to the human eye.
- Verify the output using LED observation.

- **Expected Output**

- The LED connected to the selected GPIO pin should blink continuously with a fixed ON/OFF interval.

- Refer the **Project Creation in S32 Design Studio** from S1_26_AE_AEL_DM_ZG512_Lab_CS_01.pdf





Expt.-2: Implementation of Delay Generation Using Timer (LPTMR) on the S32K144 Microcontroller.

- **Objective**

- ✓ Develop an Embedded C application using S32 SDK timer driver to generate an accurate delay using LPTMR module instead of a software busy-wait loop.

- **Problem Description**

- ✓ Configure a hardware timer module LPTMR to generate a fixed delay. Use this delay to toggle an LED or GPIO output (PTD0/PTD15/PTD16) at a known interval.

- **Functional Requirements**

- Configure system clock and peripheral clock using SDK configuration tools.
- Select LPTMR as the hardware timer source.
- Configure the timer to generate a delay of a specified duration, for example 500 ms or 1 second.
- Poll the timer status flag or use SDK blocking delay functionality.
- Toggle a GPIO output after each timer delay.
- Compare timer-based delay with software delay.

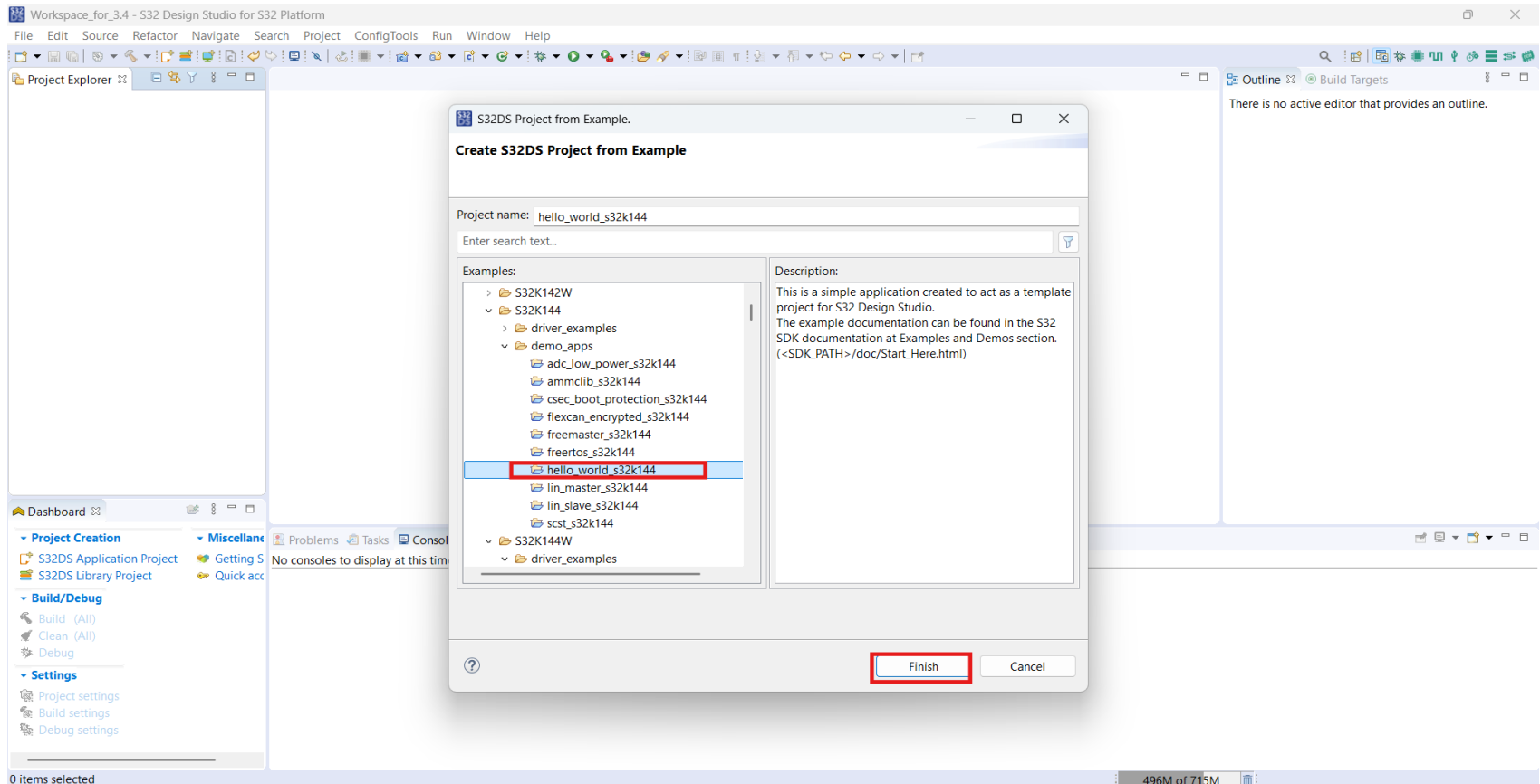
- **Expected Output**

- The GPIO pin or LED should toggle at a stable delay interval generated by LPTMR.

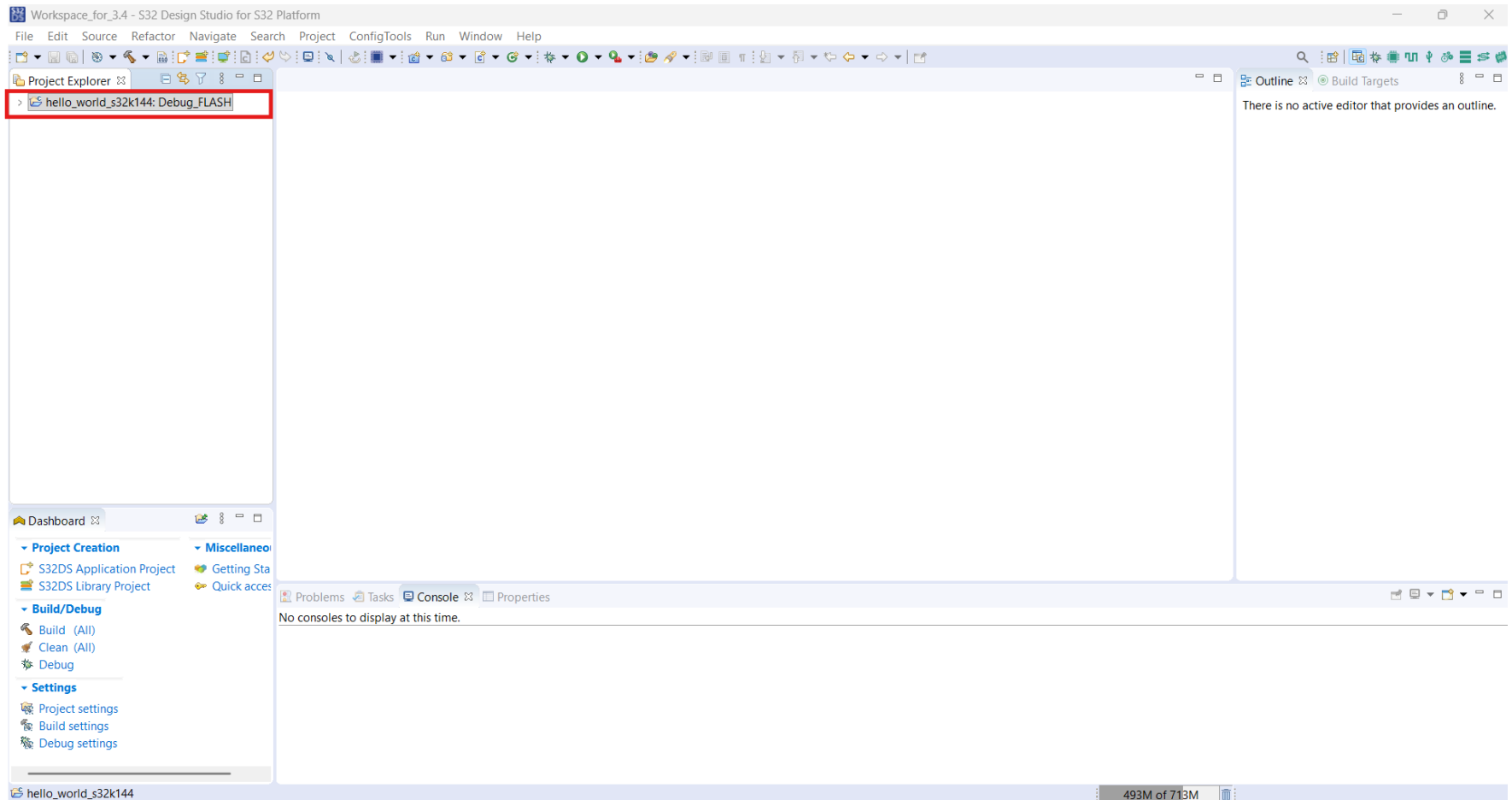
Applications of LPTMR

- Ultra-Low-Power Wake-Up Timing: Serving as a periodic system heartbeat to wake the CPU from deep sleep modes (e.g., VLPS) at defined intervals to sample sensors while keeping active power draw minimal.
- Asynchronous Pulse / Event Counting: Tracking external asynchronous pulses (such as flow meter contact closures, utility meter rotations, or button pushes) without waking the MCU core.
- ADC / Peripheral Hardware Pre-Triggering: Generating cyclic trigger strobes via TRGMUX to initiate periodic ADC conversions or comparator round-robin sequences in low-power modes.

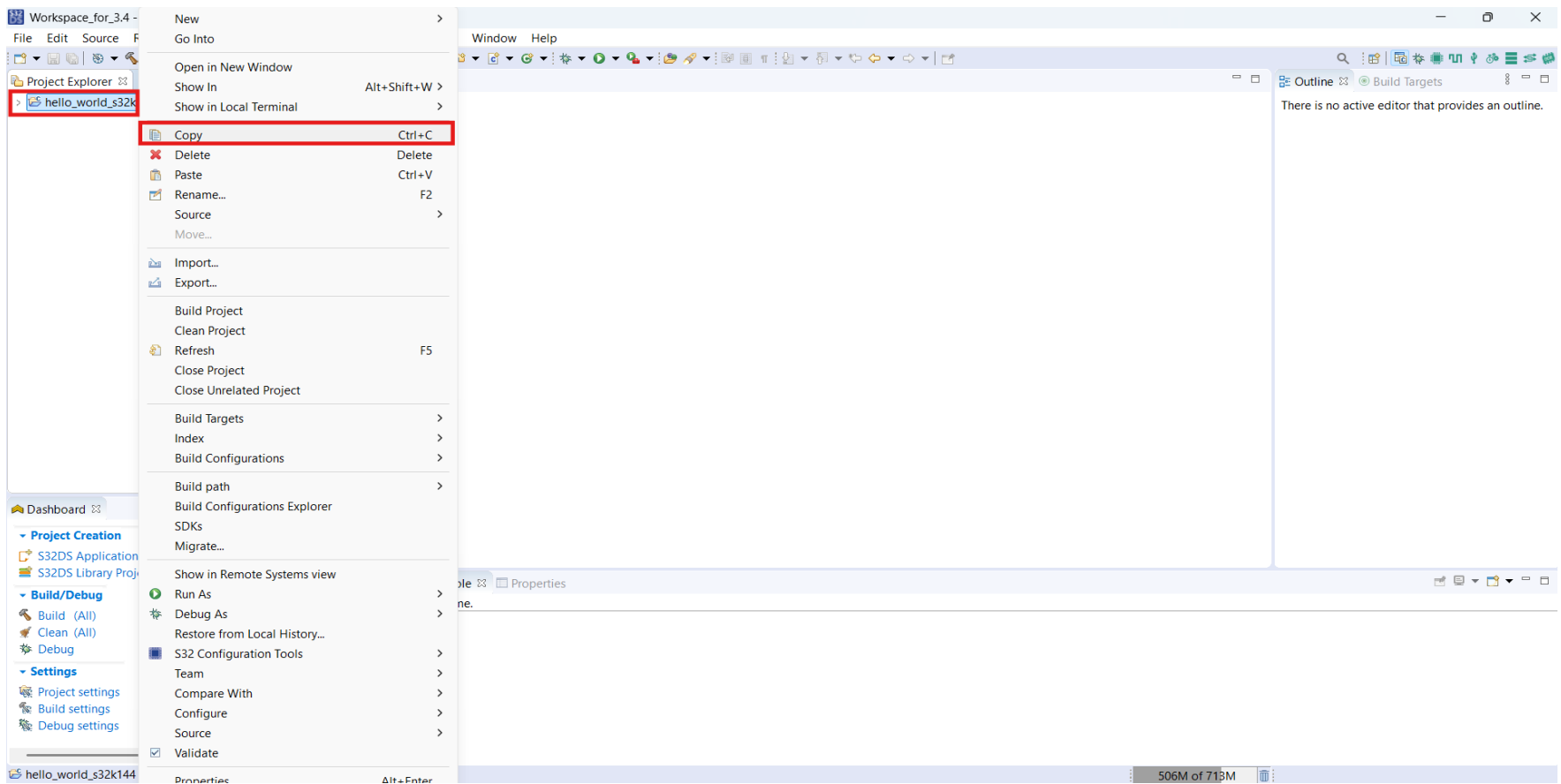
Select `hello_world_s32k144` example and click on Finish. This Project blinks the on board RGB LED which is attached at pin PD15 and PD16 of the S32K144 MCU



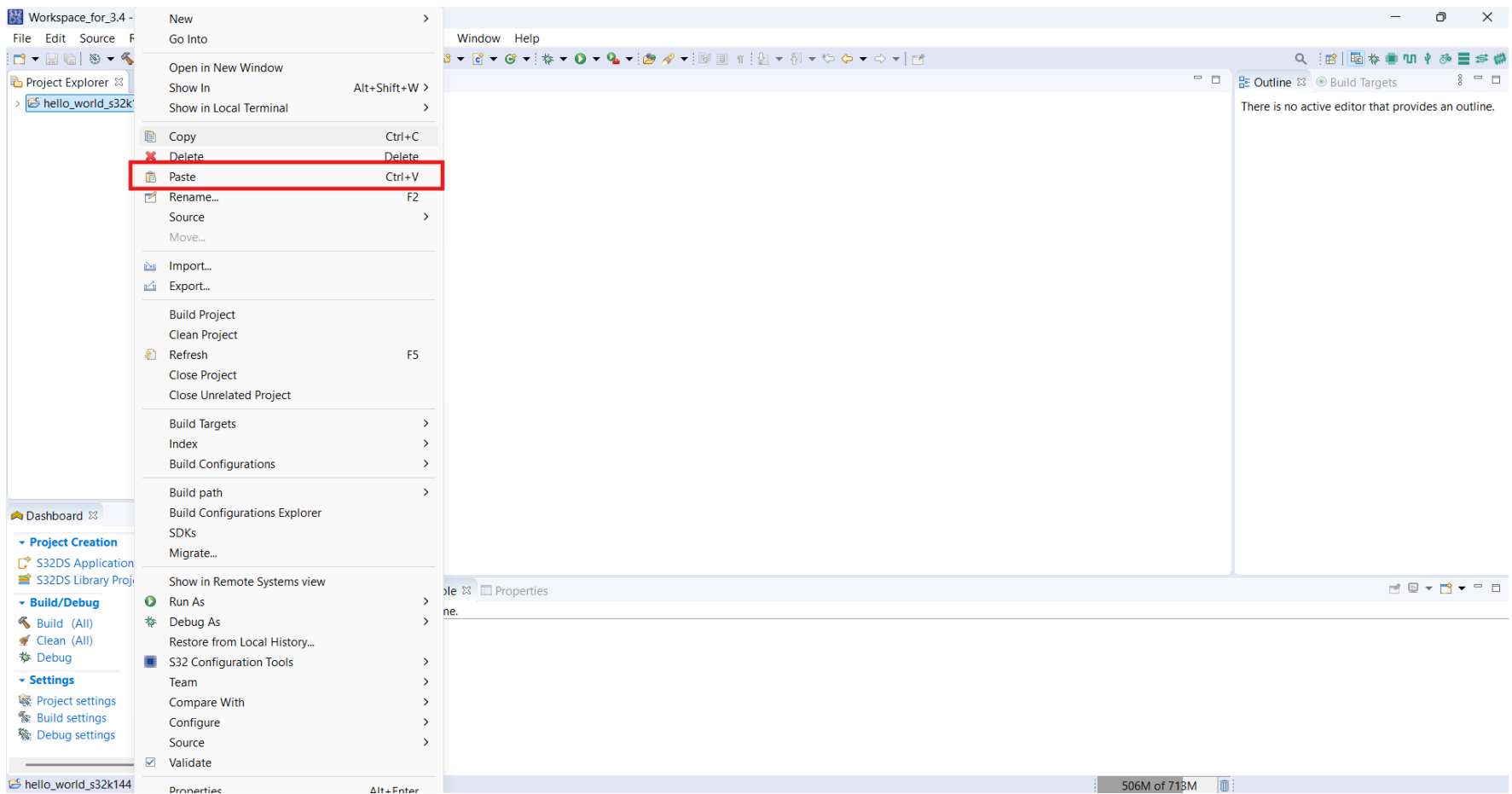
- Project is available in Project Explorer



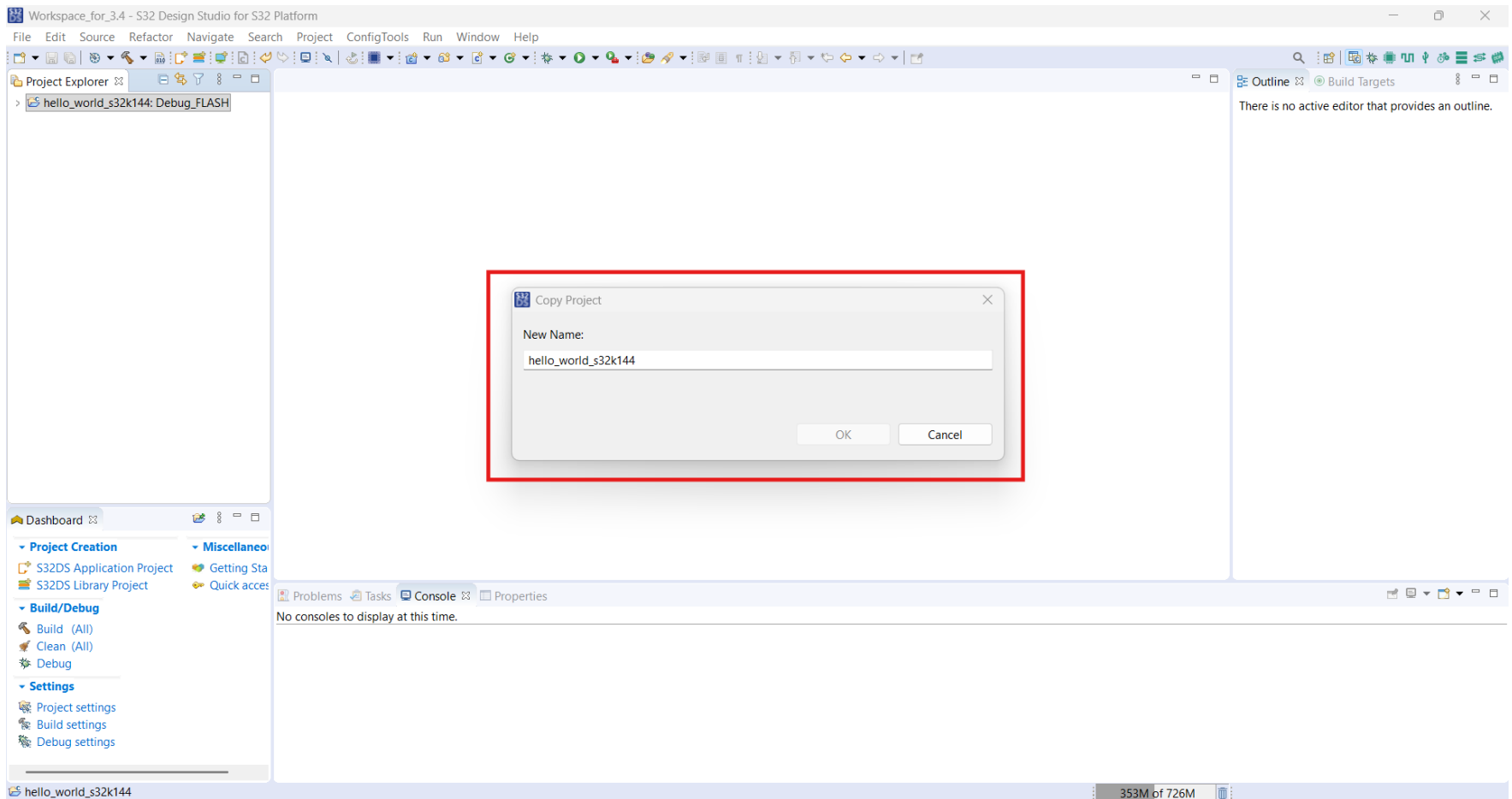
- Select hello_world_s32k144 project and right click and select **Copy**



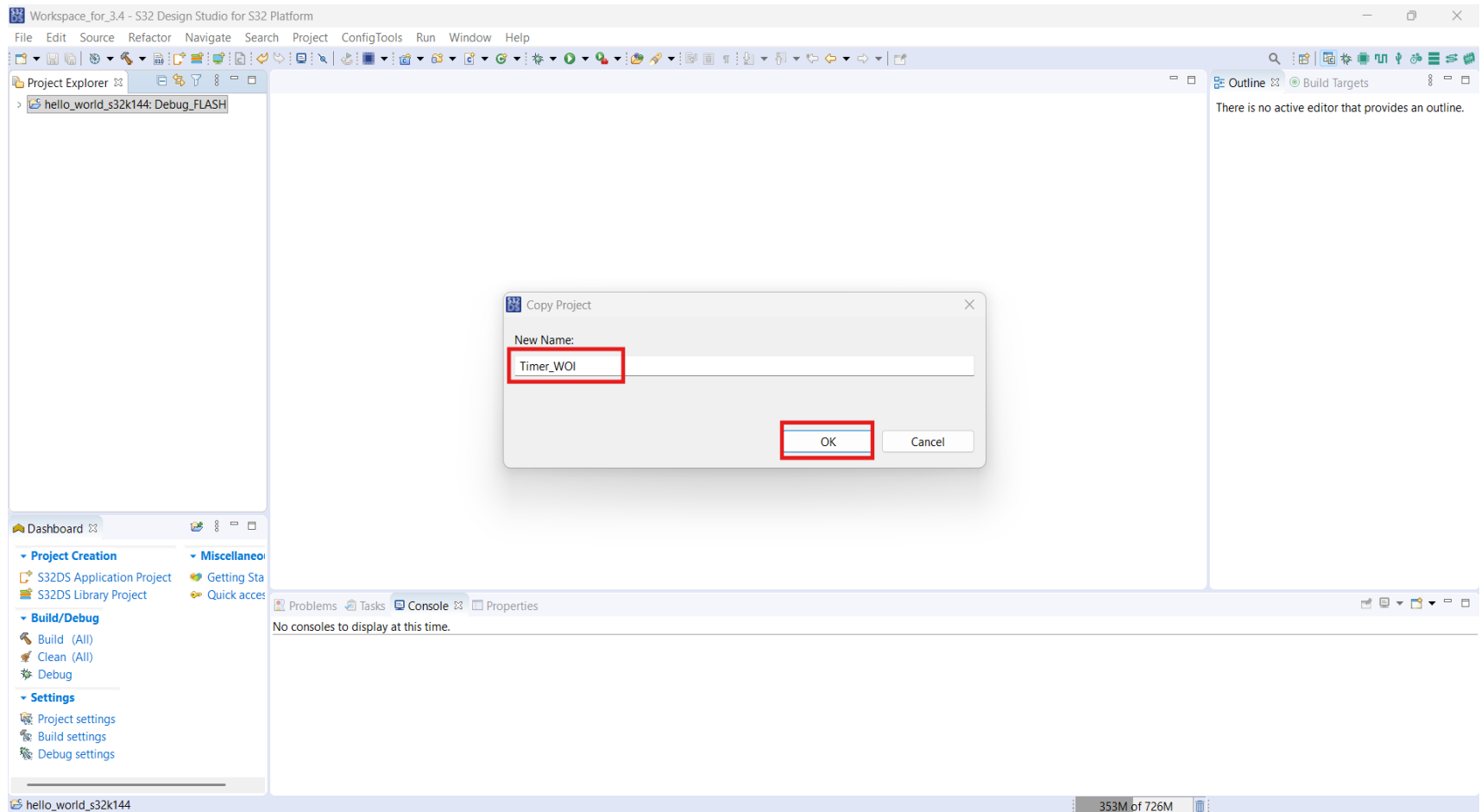
- Now click on **Paste**



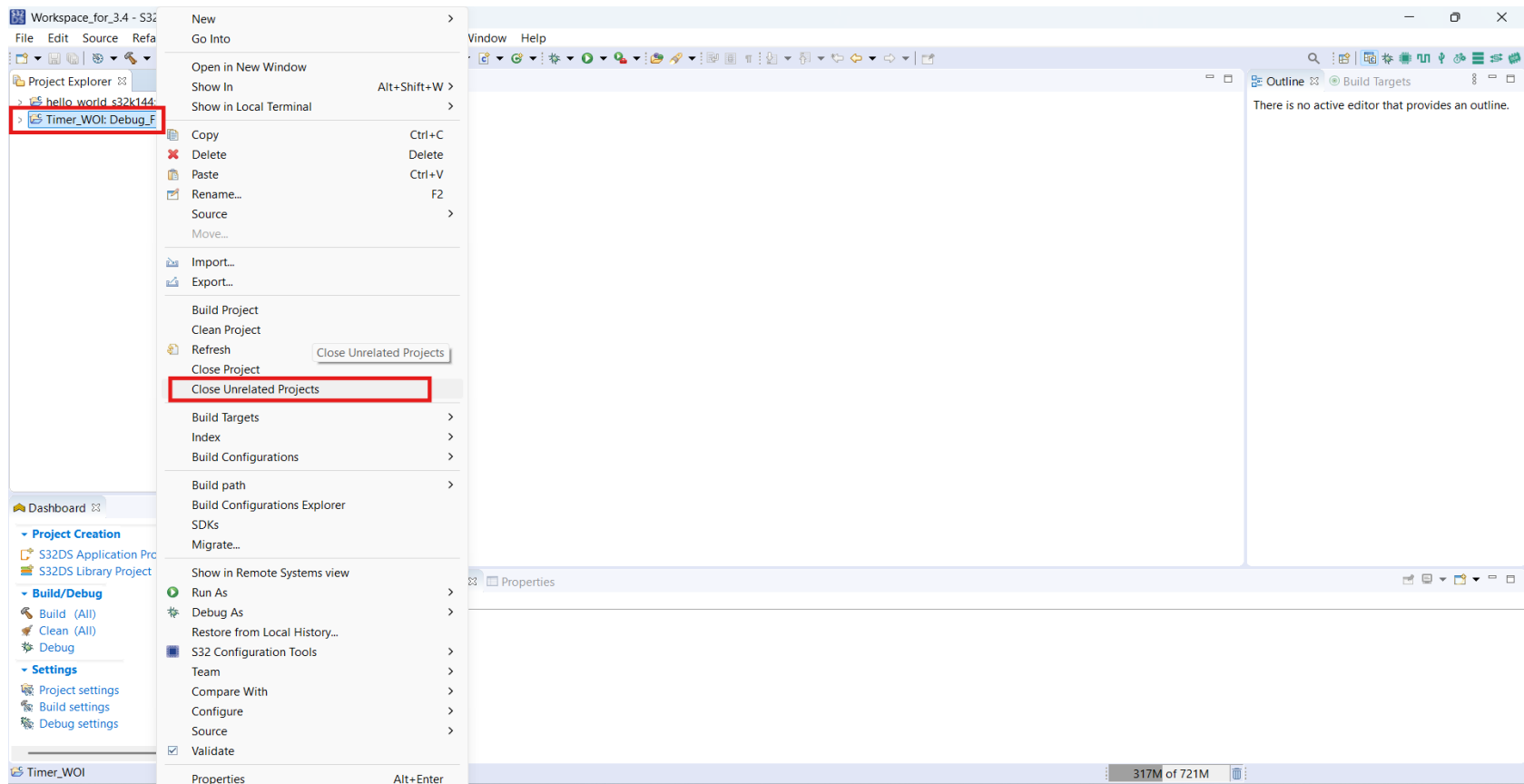
- Pop up window opens



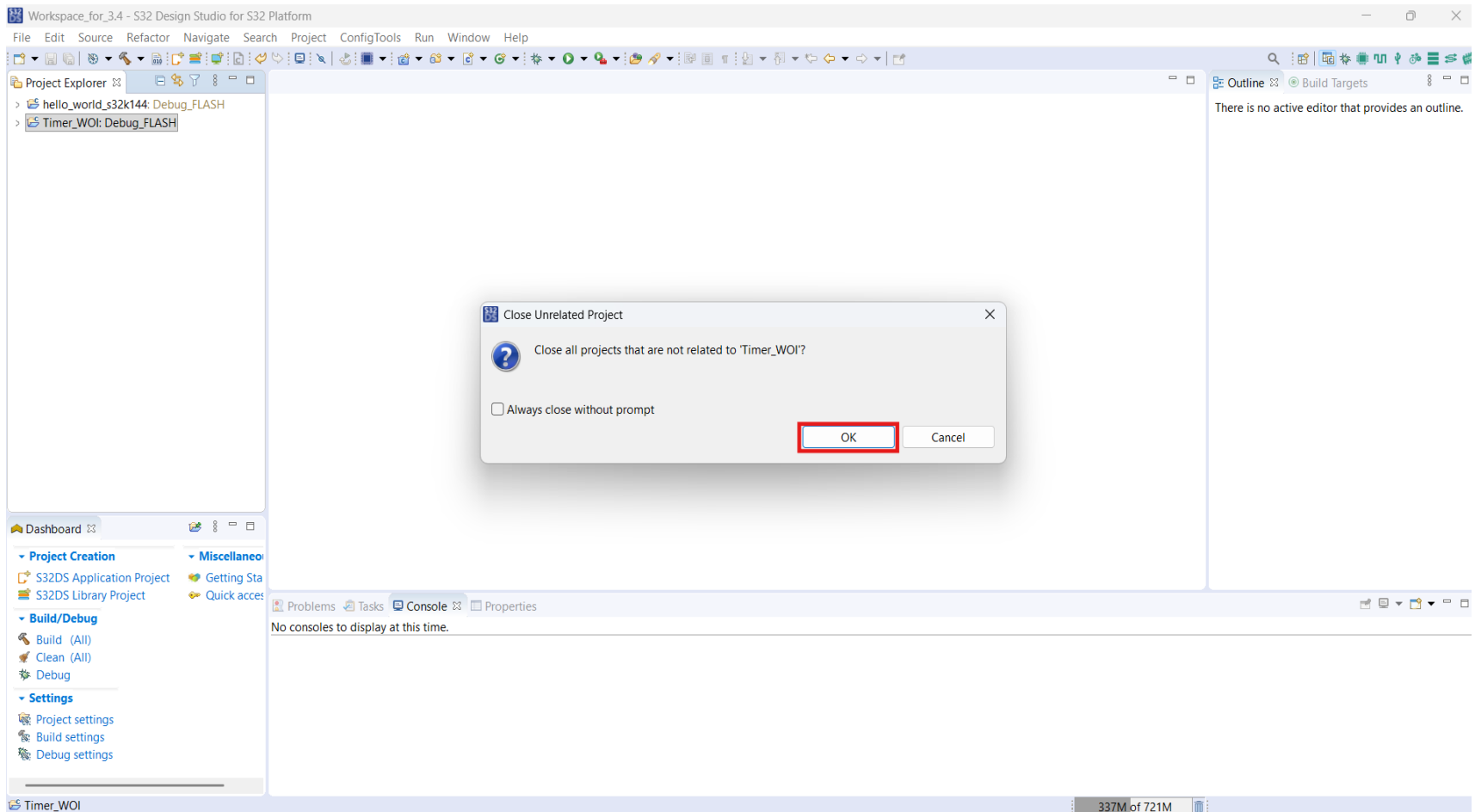
- Give name to project as Timer_WOI and Click on OK

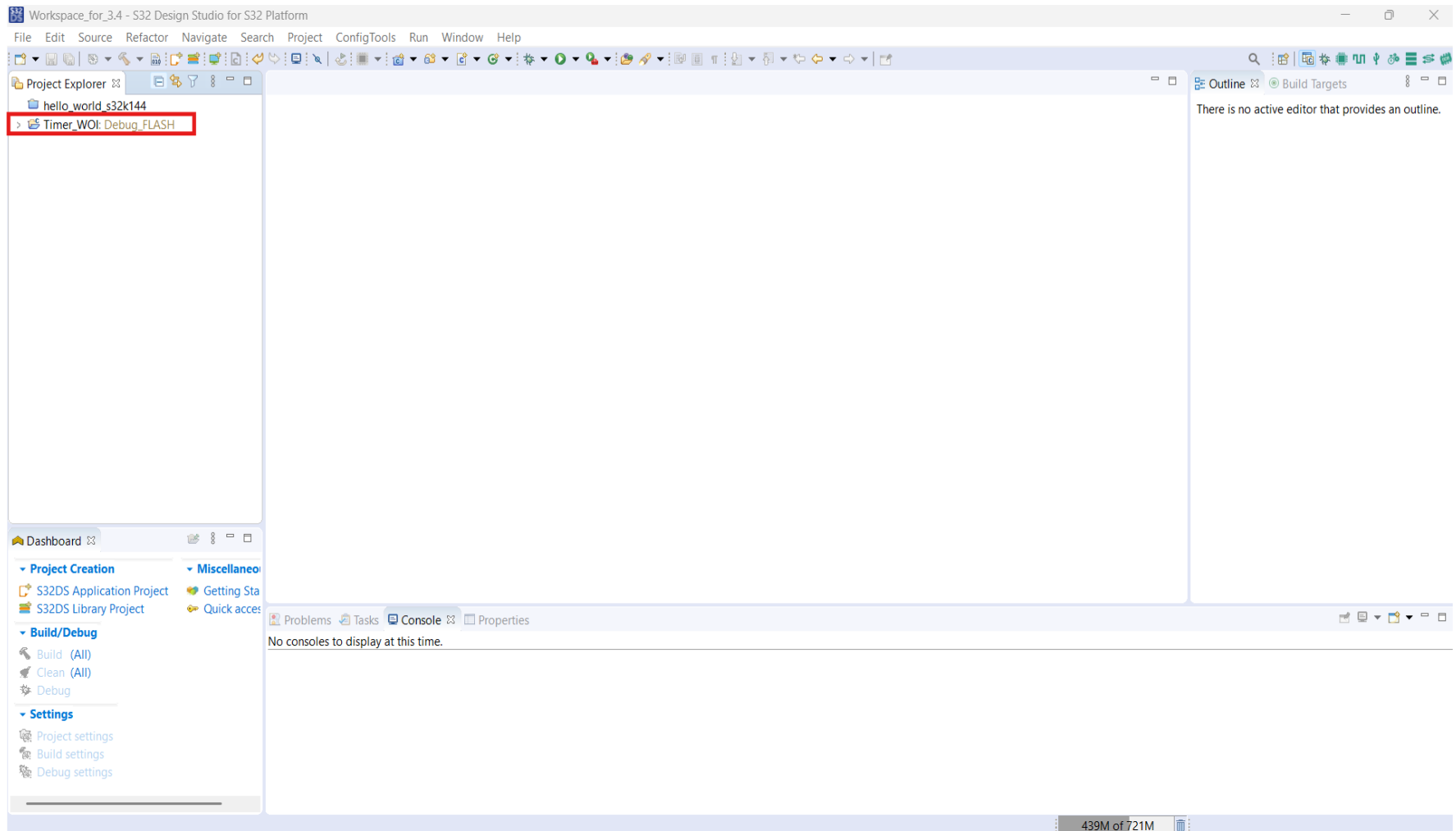


- Select Timer_WOI project and right click and click on **Close Unrelated Projects**

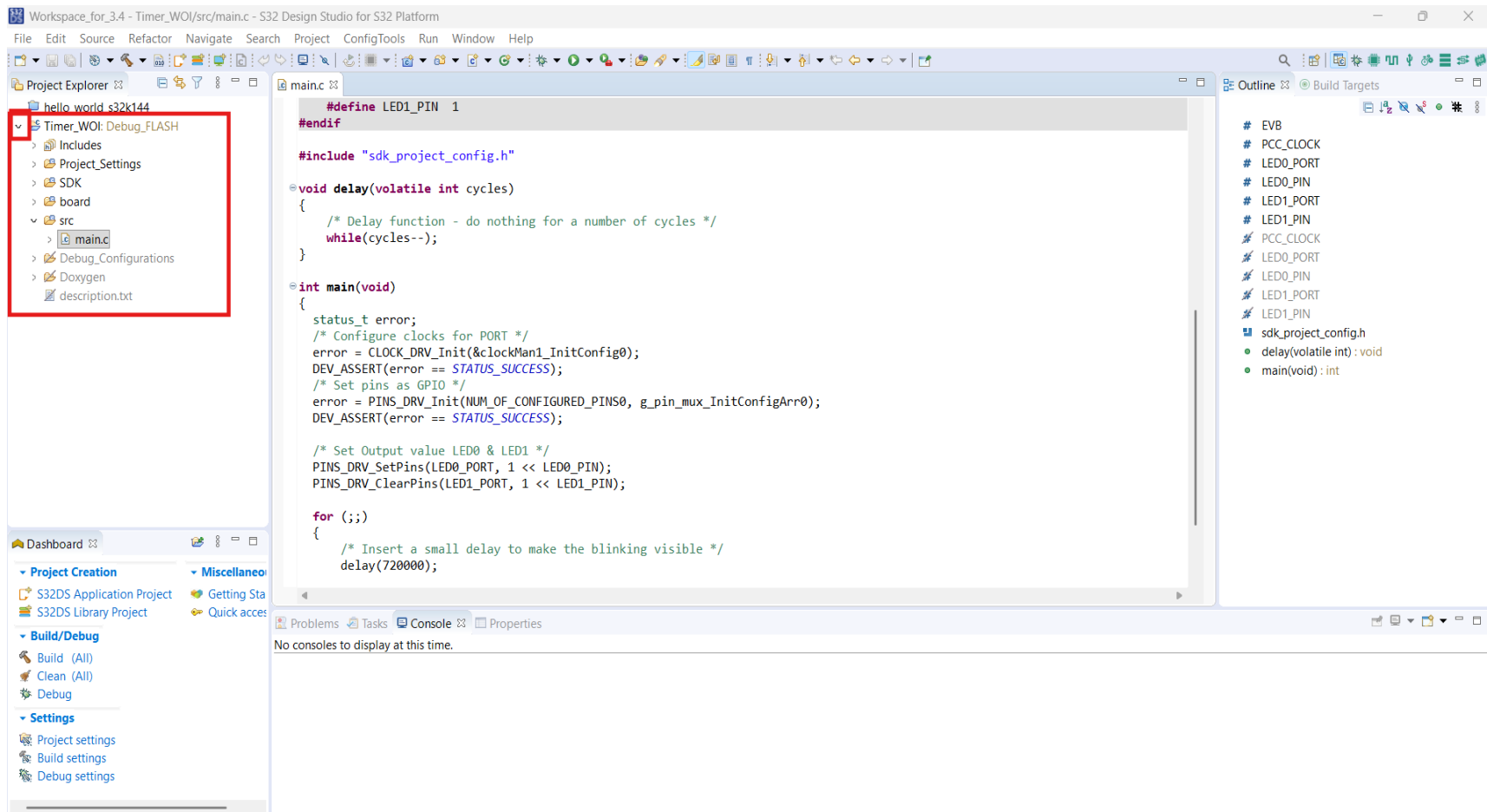


- Click on OK

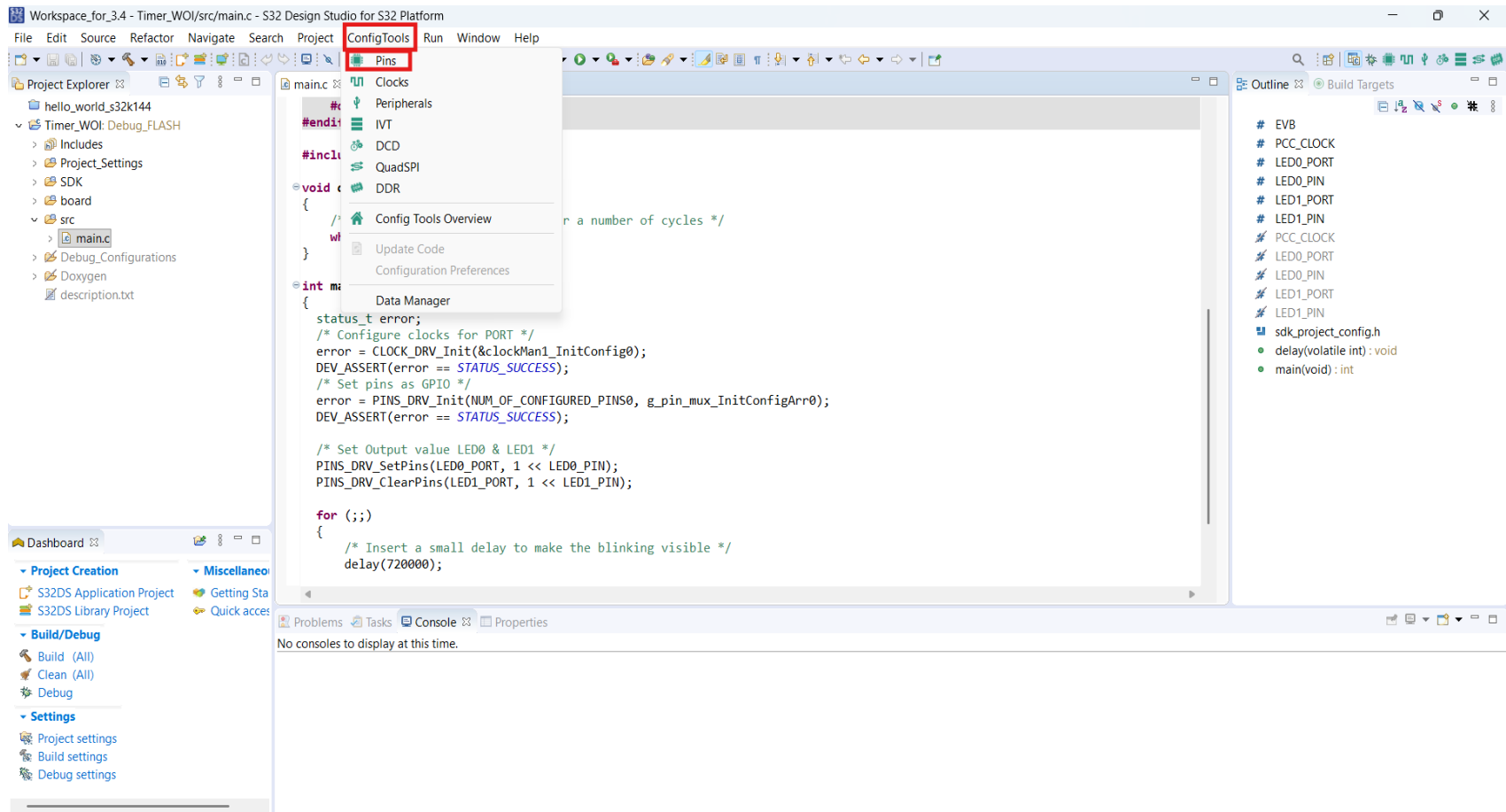




- Expand the project and open src→main.c file



- For Pin Routing Details: Click on ConfigTools→Pins



- Check your project selected

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

Timer_WOI Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

type filter text

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPi	ADC
☐	1	PTE16		PTE16	FTM2_CH7	LPSPi2_SIN	
☐	2	PTE15		PTE15	FTM2_CH6	LPSPi2_SCK	
☐	3	PTD1		PTD1	FTM0_CH3[...]	LPSPi1_SIN	
☐	4	PTD0		PTD0	FTM0_CH2[...]	LPSPi1_SCK	
☐	5	PTE11		PTE11	FTM2_CH5	LPSPi2_PCS0	
☐	6	PTE10		PTE10	FTM2_CH4	LPSPi2_PCS1	
☐	7	PTE13		PTE13	FTM2_FLT0	LPSPi2_PCS2	
☐	8	PTE5		PTE5	TCLK2[...]		
☐	9	PTE4		PTE4	FTM2_QD_P...		
☒	10	VDD_10					
☒	11	VDDA					
☒	12	VREFH					
☒	13	VREFL					
☒	14	VSS_14					
☐	15	PTB7		PTB7			
☐	16	PTB6		PTB6			
☐	17	PTE14		PTE14	FTM0_FLT1[...]		
☐	18	PTE3		PTE3	FTM0_FLT0[...]		
☐	19	PTE12		PTE12	FTM0_FLT3		
☐	20	PTD17		PTD17	FTM0_FLT2		

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 4

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
22	PORTD	port, 15	->	[22] PTD15		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
21	PORTD	port, 16	->	[21] PTD16		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled

Overview Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue

- Click on Pin 4, pop up window Pin[4] opens

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

Timer_WOI Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

Pin 4 is selected in the Pins list.

Pin [4] dialog box shows the following signals on pin 4:

- ☐ FTM0_CH2 (FTM0,ch,2)
- ☐ FTM2_CH0 (FTM2,ch,0)
- ☐ FXIO_D0 (FLEXIO,fxio_d,0)
- ☐ LPSPI1_SCK (LPSPI1,sck,sck)
- ☒ PTD0 (PORTD,port,0)
- ☐ TRGMUX_OUT1 (TRGMUX,out,1)

Done

Routing Details for BOARD... 4

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
22	PORTD	port, 15	->	[22] PTD15		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
21	PORTD	port, 16	->	[21] PTD16		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled

Overview Configuration - General Info Configuration - HW Info

Processor: S32K144
Part number: S32K144_LQFP100
Core: Cortex-M4
SDK Version: s32sdk_s32k1xx_rtm_403

Project Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

☒ Update code enabled

board\pin_mux.c
board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue

- Click on PTD0(PORTD) of Pin[4] window, pop-up Direction required window opens

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

Timer_WOI Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups

Pin [4]

All signals on pin [4]:

- ☐ FTM0_CH2 (FTM0,ch,2)
- ☐ FTM0_CH0 (FTM0,ch,0)
- ☐ FXIO_D0 (FLEXIO,fxio_d,0)
- ☐ LPSP11_SCK (LPSP11,sck,sck)
- ☒ PTD0 (PORTD,port,0)
- ☐ TRGMUX_OUT1 (TRGMUX,out,1)

Direction required!

Select the routing direction for:
[Pin: PTD0 | Peripheral: PORTD | Signal: port, 0]

Input
Output

Done

OK Cancel

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 4

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
22	PORTD	port, 15	->	[22] PTD15		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
21	PORTD	port, 16	->	[21] PTD16		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled

Timer_WOI

388M of 1030M

Overview Code Preview Registers

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue

- Select Output of Direction required window and Click OK

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

Timer_WOI Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups

Pin type filter text

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPI	ADC
1	PTE16			PTE16	FTM2_CH7	LPSPI2_SIN	
2	PTE15			PTE15	FTM2_CH6	LPSPI2_SCK	
3	PTD1			PTD1	FTM0_CH3[...]	LPSPI1_SIN	
4	PTD0			PTD0	FTM0_CH2[...]	LPSPI1_SCK	
5	PTE11			PTE11	FTM2_CH5	LPSPI2_PCS0	
6	PTE10			PTE10	FTM2_CH4	LPSPI2_PCS1	
7	PTE13			PTE13	FTM2_FLT0	LPSPI2_PCS2	
8	PTE5			PTE5	TCLK2[...]		
9	PTE4			PTE4	FTM2_QD_P...		
10	VDD_10						
11	VDDA						
12	VREFH						
13	VREFL						
14	VSS_14						
15	PTB7			PTB7			
16	PTB6			PTB6			
17	PTE14			PTE14	FTM0_FLT1[...]		
18	PTE3			PTE3	FTM0_FLT0[...]		
19	PTE12			PTE12	FTM0_FLT3		
20	PTD17			PTD17	FTM0_FLT2		

Routing Details

Pins Signals type filter text

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter
40	PORTC	port_0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
39	PORTC	port_1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
22	PORTD	port_15	->	[22] PTD15		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
21	PORTD	port_16	->	[21] PTD16		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled

Pin [4]

All signals on pin [4]:

- ☐ FTM0_CH2 (FTM0,ch,2)
- ☐ FTM2_CH0 (FTM2,ch,0)
- ☐ FXIO_D0 (FLEXIO,fxio_d,0)
- ☐ LPSPI1_SCK (LPSPI1,sck,sck)
- ☒ PTD0 (PORTD,port,0)
- ☐ TRGMUX_OUT1 (TRGMUX,out,1)

Done

Direction required!

Select the routing direction for:
[Pin: PTD0 | Peripheral: PORTD | Signal: port_0]

Input
Output

OK Cancel

Overview Configuration - General Info Configuration - HW Info Processor: S32K144 Part number: S32K144_LQFP100 Core: Cortex-M4 SDK Version: s32sdk_s32k1xx_rtm_403 Project Pins Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration. Generated code Update code enabled board/pin_mux.c board/pin_mux.h Functional groups BOARD_InitPins Other tools Problems type filter text Level Resource Issue

- Click on Done

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

Timer_WOI Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups

Pin Pin name Label Identifier PORT FTM LPSPI ADC

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPI	ADC
1	PTE16			PTE16	FTM2_CH7	LPSPI2_SIN	
2	PTE15			PTE15	FTM2_CH6	LPSPI2_SCK	
3	PTD1			PTD1	FTM0_CH3[...]	LPSPI1_SIN	
4	PTD0			PTD0	FTM0_CH2[...]	LPSPI1_SCK	
5	PTE11			PTE11	FTM2_CH5	LPSPI2_PCS0	
6	PTE10			PTE10	FTM2_CH4	LPSPI2_PCS1	
7	PTE13			PTE13	FTM2_FLT0	LPSPI2_PCS2	
8	PTE5			PTE5	TCLK2[...]		
9	PTE4			PTE4	FTM2_QD_P...		
10	VDD_10						
11	VDDA						
12	VREFH						
13	VREFL						
14	VSS_14						
15	PTB7		PTB7				
16	PTB6		PTB6				
17	PTE14		PTE14		FTM0_FLT1[...]		
18	PTE3		PTE3		FTM0_FLT0[...]		
19	PTE12		PTE12		FTM0_FLT3		
20	PTD17		PTD17		FTM0_FLT2		

Package

Pin [4]

All signals on pin [4]:

- ☐ FTM0_CH2 (FTM0.ch,2)
- ☐ FTM2_CH0 (FTM2.ch,0)
- ☐ FXIO_D0 (FLEXIO.fxio_d,0)
- ☐ LPSPI1_SCK (LPSPI1.sck,sck)
- ☒ PTD0 (PORTD.port,0)
- ☐ TRGMUX_OUT1 (TRGMUX.out,1)

Done

Routing Details

Pins Signals

Routing Details for BOARD... 5

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
22	PORTD	port, 15	->	[22] PTD15		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
21	PORTD	port, 16	->	[21] PTD16		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled
4	PORTD	port, 0	->	[4] PTD0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled

Overview Configuration - General Info Configuration - HW Info Project Pins Generated code Functional groups Other tools Problems

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue

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- Give Label, Identifier and Initial value as shown below:

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

Timer_WOI Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

Pin name Label Identifier PORT FTM

1	PTE16		PTE16	FTM2_CH7
2	PTE15		PTE15	FTM2_CH6
3	PTD1		PTD1	FTM0_CH3[...]
4	PTD0	RGB_LED_BLUE	RGB_LED_BLUE	FTM0_CH2[...]
5	PTE11		PTE11	FTM2_CH5
6	PTE10		PTE10	FTM2_CH4
7	PTE13		PTE13	FTM2_FLT0
8	PTE5		PTE5	TCLK2[...]
9	PTE4		PTE4	FTM2_QD_P...
10	VDD_10			
11	VDDA			
12	VREFH			
13	VREFL			
14	VSS_14			
15	PTB7		PTB7	
16	PTB6		PTB6	
17	PTE14		PTE14	FTM0_FLT1[...]
18	PTE3		PTE3	FTM0_FLT0[...]
19	PTE12		PTE12	FTM0_FLT3
20	PTD17		PTD17	FTM0_FLT2

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 5

Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter	Drive Strength	Passive Filter	Initial Value
->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	n/a	n/a	Low
->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	n/a	n/a	Low
->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	Low	n/a	High
->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	Low	n/a	High
->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	Low	n/a	High

S32K144_LQFP100 - LQFP 100 package

Configuration - C

Configuration - F

Processor: S32K1

Part number: S32K1

Core: Cortex

SDK Version: s32sdl

Project

Pins

Config electric and ru

Generated code

Update code ena

board\pin_mux.

board\pin_mux.

Functional group

BOARD_InitPins

Other tools

Problems

type filter text

Level Resource

- For Peripheral Configuration: Select ConfigTools→Peripherals

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project **ConfigTools** Pins Run Window Help

Timer_WOI BOARD_InitPins

Pins Peripheral... Power Gro... Package

ConfigTools menu:

- Pins
- Clocks
- Peripherals**
- IVT
- DCD
- QuadSPI
- DDR
- Config Tools Overview
- Update Code
- Configuration Preferences
- Data Manager

Pin list:

Pin	Pin name	Label	Identifier	PO
1	PTE16			PT16
2	PTE15			PT15
3	PTD1			PTD1
4	PTD0	RGB_LE...	RGB_LED_...	PTD0
5	PTE11			PT11
6	PTE10			PT10
7	PTE13			PT13
8	PTE5			PT5
9	PTE4			PT4
10	VDD_10			
11	VDDA			
12	VREFH			
13	VREFL			
14	VSS_14			
15	PTB7			PTB7
16	PTB6			PTB6
17	PTB4			PTB4
18	PTE3			PT3
19	PTE12			PT12

Routing Details for BOARD...

Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Sel
->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do

Overview Configuration - General Info

Configuration - HW Info

Processor: S32K144
Part number: S32K144_LQFP100
Core: Cortex-M4
SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c
board\pin_mux.h

Functional groups

BOARD_InitPins

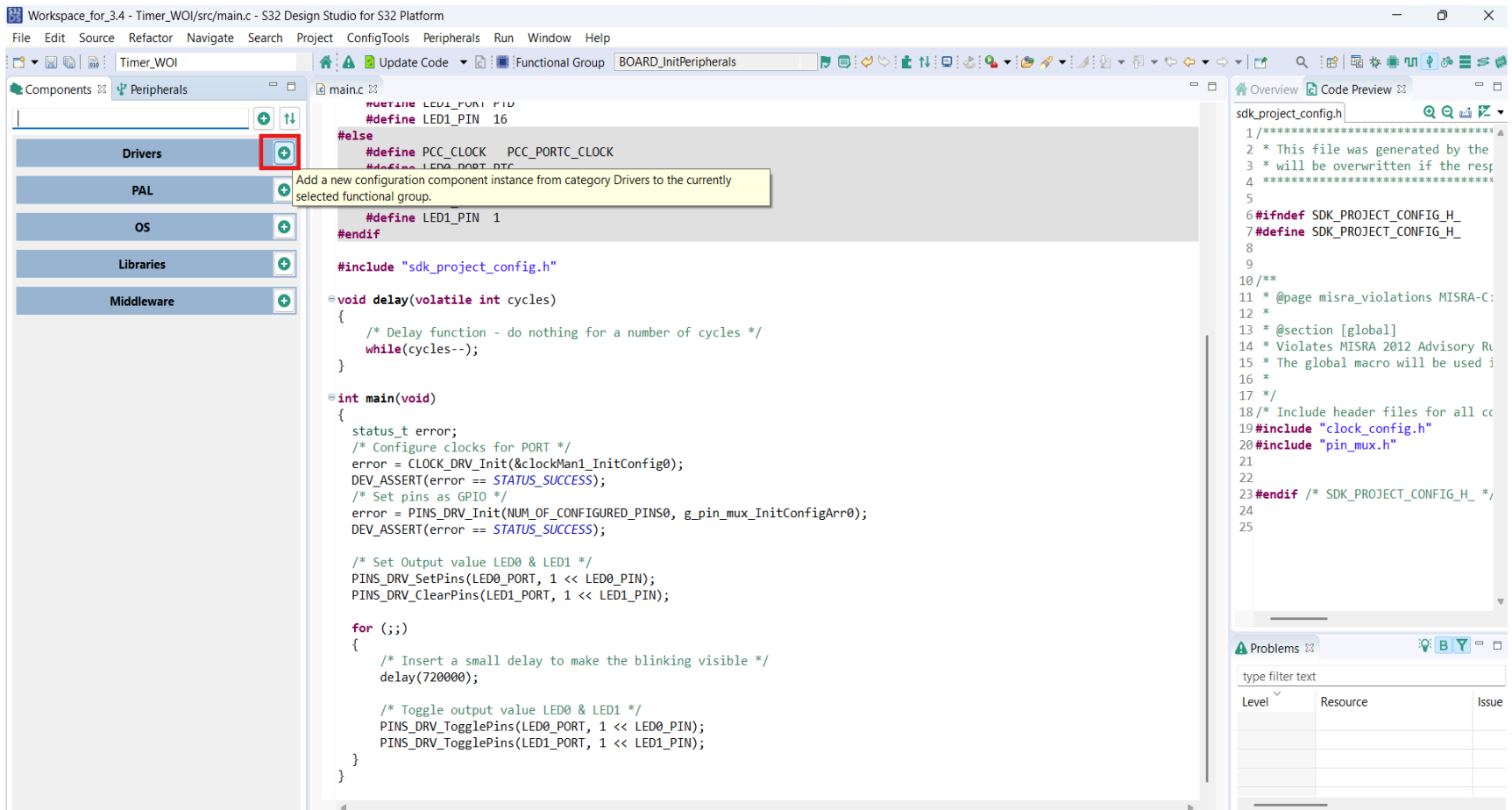
Other tools

Problems

type filter text

Level	Resource	Issue	Origin

- Click on Add Drivers, new pop-up window opens



- Select **lptmr** and Click on OK

The screenshot displays the S32 Design Studio interface. In the center, a 'Select configuration component' dialog box is open, showing a list of configuration components. The 'lptmr' component, described as 'Low Power Timer', is highlighted with a red box. Below the list, the 'OK' button is also highlighted with a red box. The background shows the 'main.c' file with the following code:

```
#define LED0_PORT 16
#define LED1_PORT 16

/* Delay function */
void delay(volatile unsigned int cycles)
{
    while(cycles-- > 0)
    {
        /* Delay function */
    }
}

int main(void)
{
    status_t error;
    /* Configure clock */
    error = CLOCK_DRV_Init(CLOCK_DRV_INIT_FLAGS_NO_INIT_FLAGS, NULL, NULL);
    DEV_ASSERT(error == 0);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(1, NULL, NULL);
    DEV_ASSERT(error == 0);

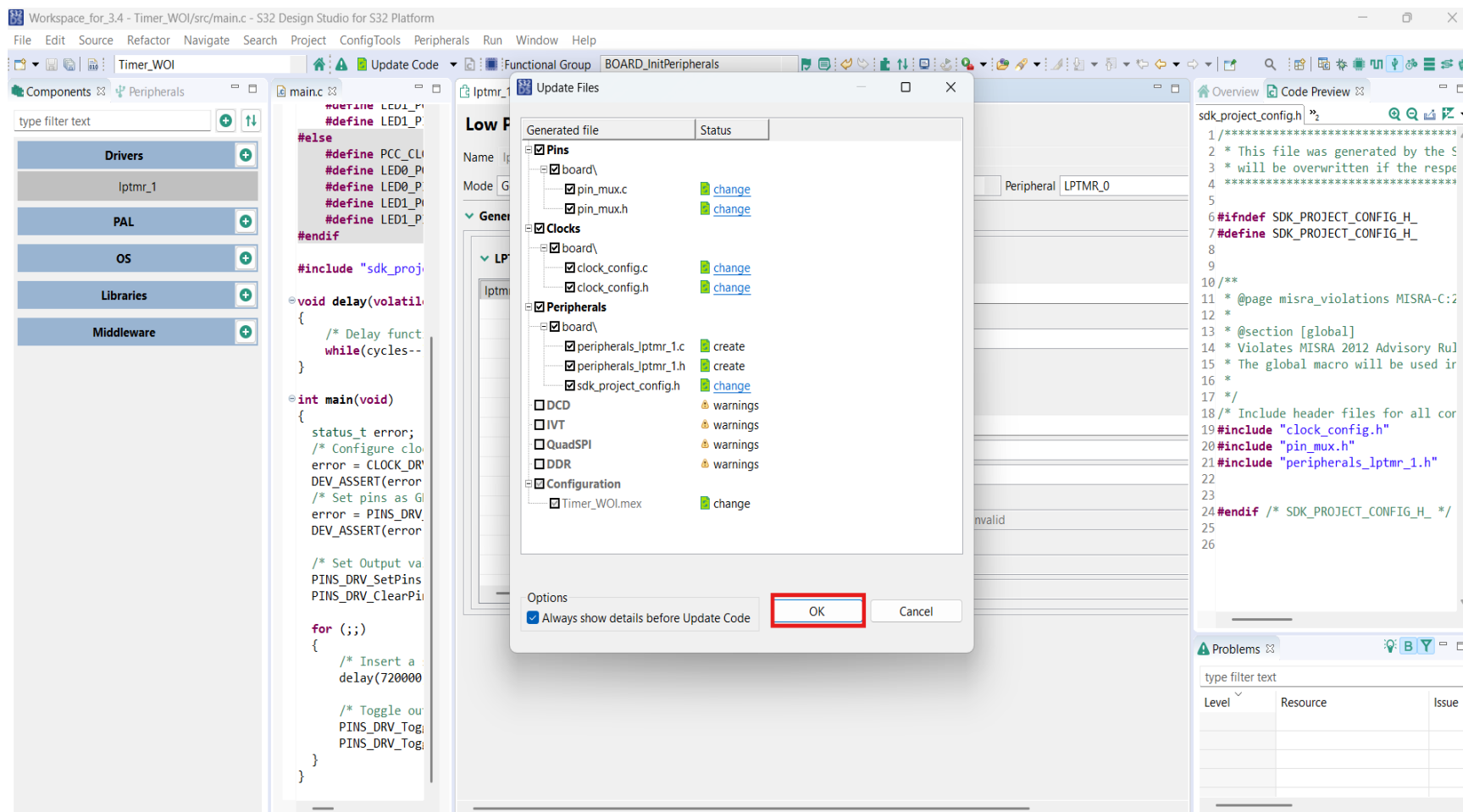
    /* Set Output value LED0 & LED1 */
    PINS_DRV_SetPins(LED0_PORT, 1 << LED0_PIN);
    PINS_DRV_ClearPins(LED1_PORT, 1 << LED1_PIN);

    for (;;)
    {
        /* Insert a small delay to make the blinking visible */
        delay(720000);

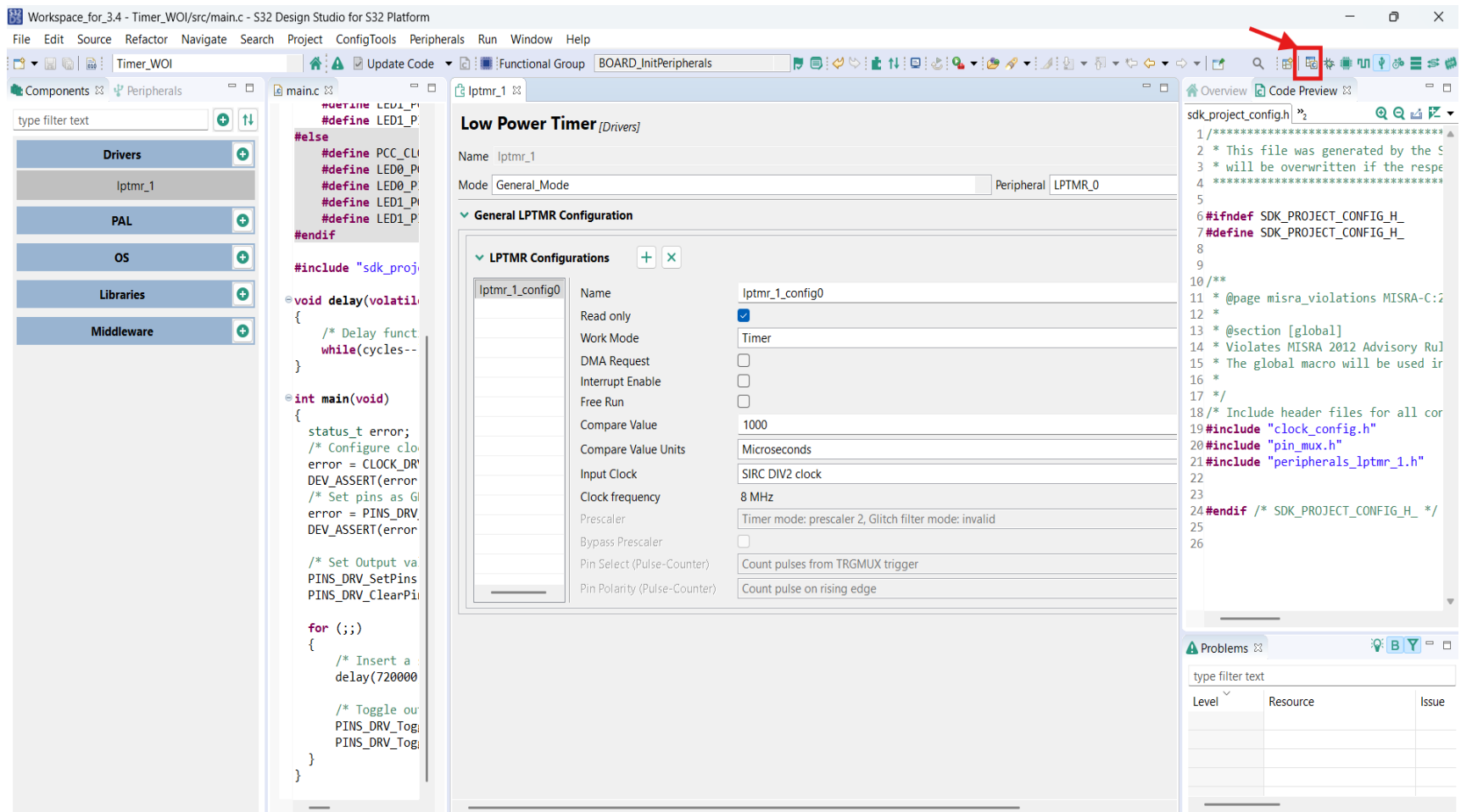
        /* Toggle output value LED0 & LED1 */
        PINS_DRV_TogglePins(LED0_PORT, 1 << LED0_PIN);
        PINS_DRV_TogglePins(LED1_PORT, 1 << LED1_PIN);
    }
}
```

The 'Components' sidebar on the left shows the 'Peripherals' section with 'Drivers', 'PAL', 'OS', 'Libraries', and 'Middleware' categories. The 'sdk_project_config.h' file is visible on the right, showing configuration macros.

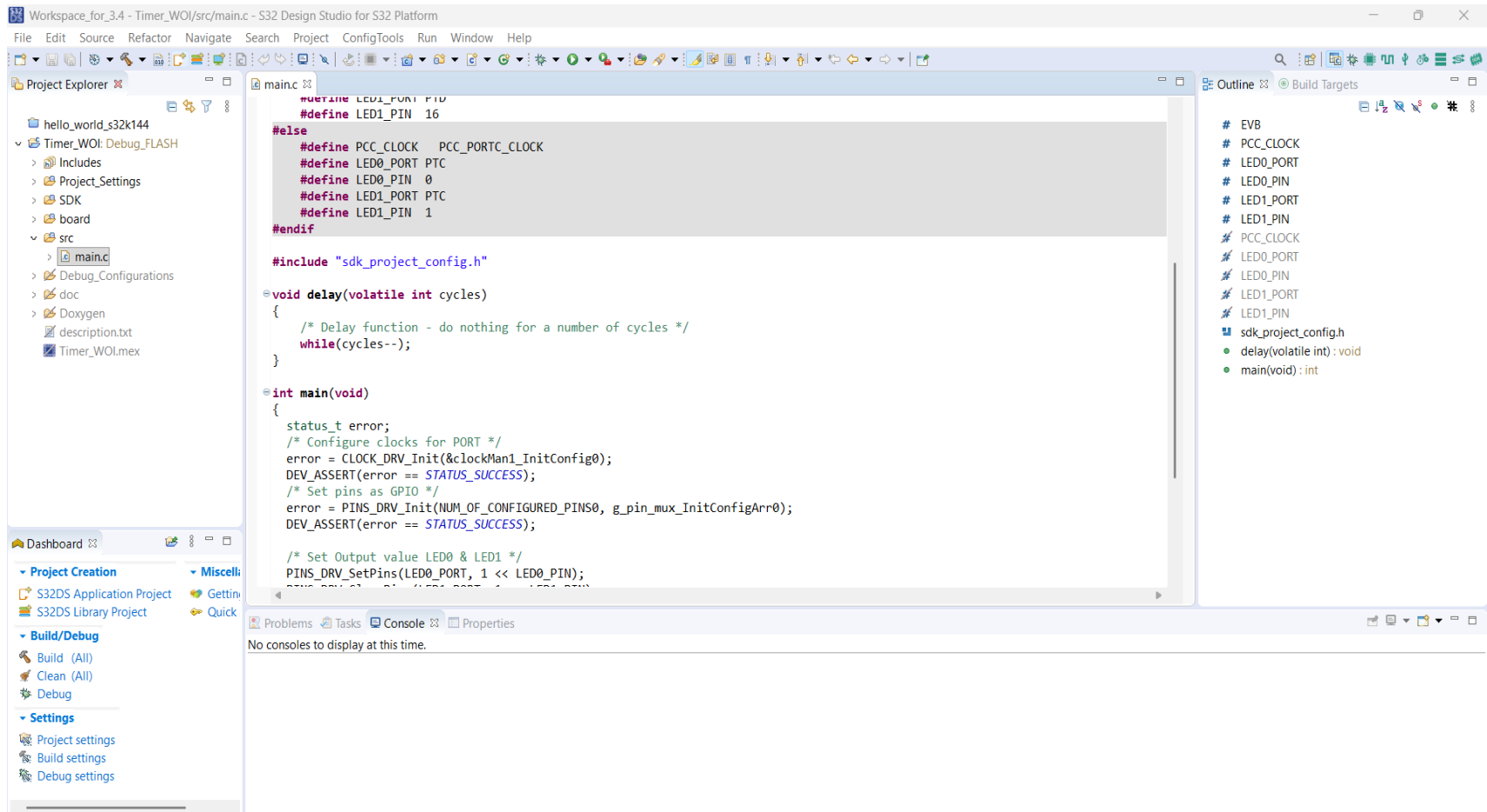
- Click on OK



- After update click on C/C++ to change perspective



- C/C++ perspective



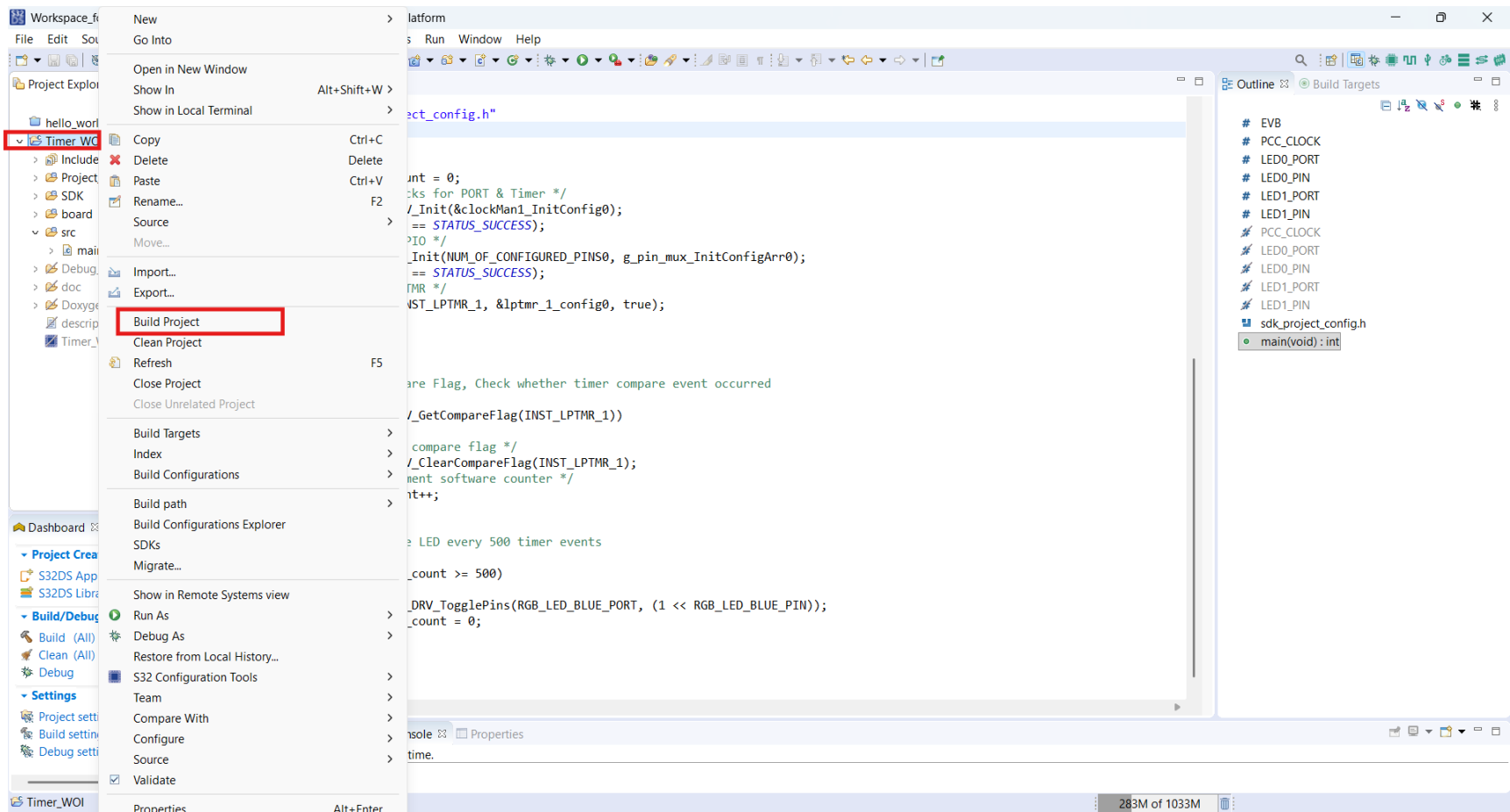
- Write C program as below and save it.

```
*main.c
#include "sdk_project_config.h"
int main(void)
{
    status_t error;
    uint16_t tick_count = 0;
    /* Configure clocks for PORT & Timer */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Initialize LPTMR */
    LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);

    while (1)
    {
        /*
         * Poll Compare Flag, Check whether timer compare event occurred
         */
        if (LPTMR_DRV_GetCompareFlag(INST_LPTMR_1))
        {
            /* Clear compare flag */
            LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);
            /* Increment software counter */
            tick_count++;

            /*
             * Toggle LED every 500 timer events
             */
            if (tick_count >= 500)
            {
                PINS_DRV_TogglePins(RGB_LED_BLUE_PORT, (1 << RGB_LED_BLUE_PIN));
                tick_count = 0;
            }
        }
    }
}
```

- Right click on project and click on Build Project



- Check for any errors

The screenshot displays the S32 Design Studio IDE interface. The main editor shows the `main.c` file with the following code:

```
#include "sdk_project_config.h"
int main(void)
{
    status_t error;
    uint16_t tick_count = 0;
    /* Configure clocks for PORT & Timer */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Initialize LPTMR */
    LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);

    while (1)
    {
        /* Poll Compare Flag, Check whether timer compare event occurred */
        if (LPTMR_DRV_GetCompareFlag(INST_LPTMR_1))
        {
            /* Clear compare flag */
            LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);
            /* Increment software counter */
            tick_count++;
        }
    }
}
```

The Project Explorer on the left shows the project structure, including the `src` directory containing `main.c`. The Outline view on the right lists the project's components, including `main(void):int`.

The Console window at the bottom, highlighted with a red box, displays the build output:

```
CDT Build Console [Timer_WOI]
Invoking: Standard S32DS C Linker
arm-none-eabi-gcc -o "Timer_WOI.elf" "@Timer_WOI.args"
Finished building target: Timer_WOI.elf

Invoking: Standard S32DS Print Size
arm-none-eabi-size --format=berkeley Timer_WOI.elf
text    data    bss    dec    hex filename
9228    576    3096    12900   3264 Timer_WOI.elf
Finished building: Timer_WOI.siz

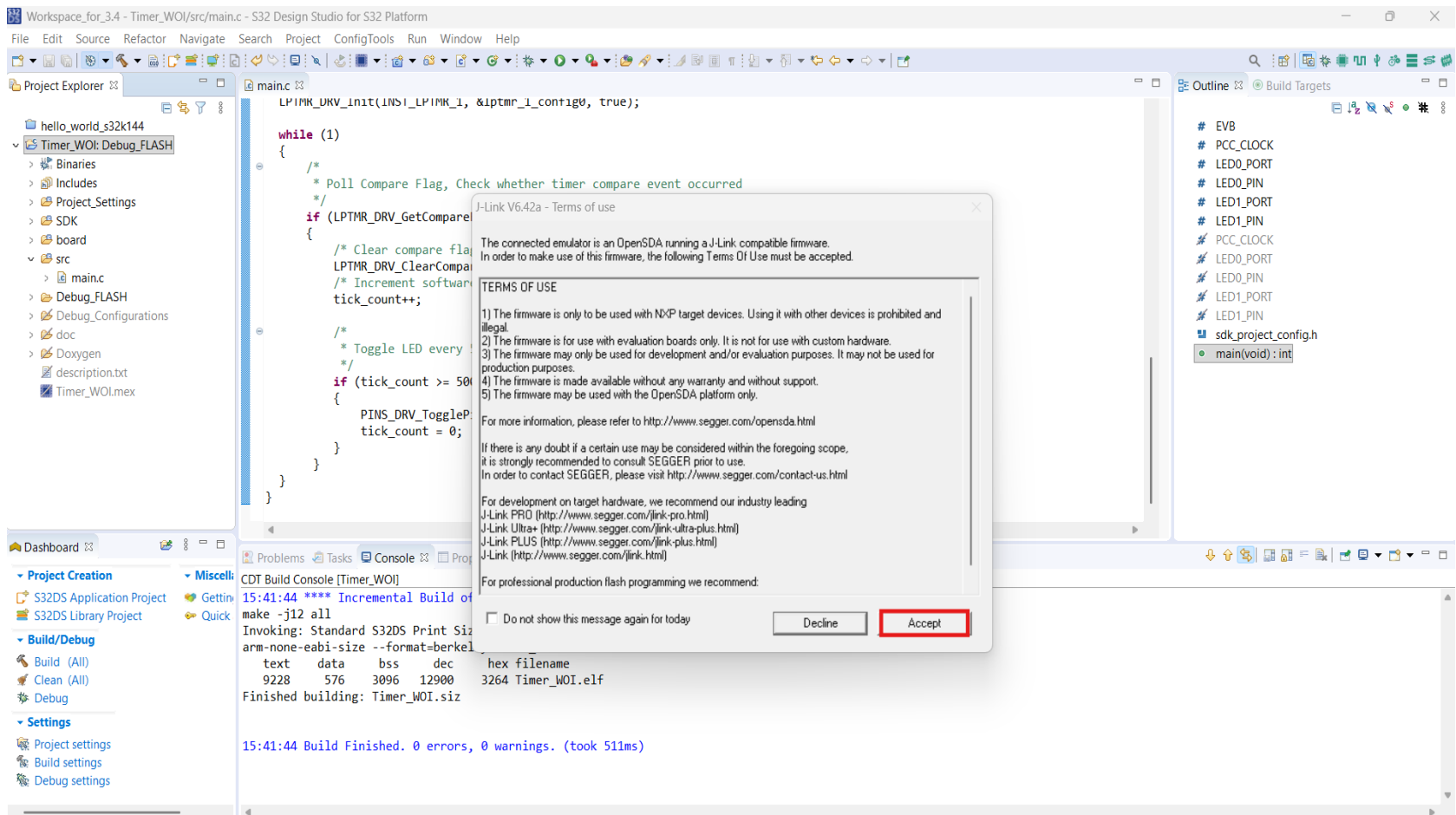
15:30:27 Build Finished. 0 errors, 0 warnings. (took 5s.814ms)
```

-
- The screenshot displays the S32 Design Studio IDE interface. The top menu bar includes File, Edit, Source, Refactor, Navigate, Search, Project, ConfigTools, Run, Window, and Help. The Project Explorer on the left shows the project structure, with 'Timer_WOI: Debug_FLASH' selected. The main editor window shows the 'main.c' source code, which includes a while loop for timer events. A context menu is open over the code, showing options like 'Debug As' and 'Debug Configurations...'. The right sidebar shows the 'Outline' view with a list of variables and functions. The bottom console shows the build output, indicating a successful build of 'Timer_WOI.elf'.

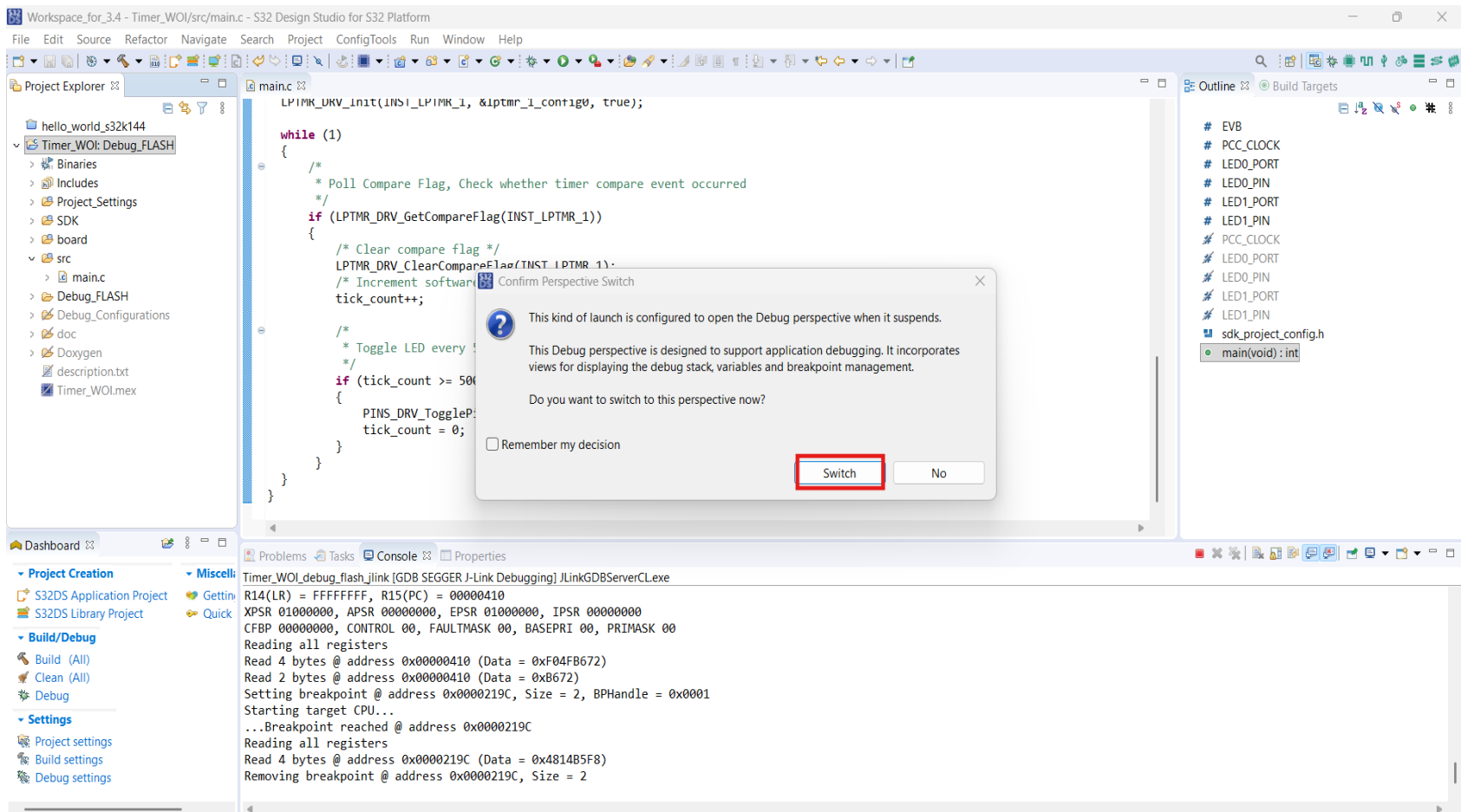
- Do configurations as below and Click on **Debug**

The screenshot shows the Eclipse IDE interface with the **Debug Configurations** dialog open. The **Project Explorer** on the left shows a project named **hello_world_s32k144** with a sub-project **Timer_WOI: Debug_FLASH**. The **Debug Configurations** dialog is titled "Create, manage, and run configurations" and lists various debuggers. The **GDB SEGGER J-Link Debugging** category is expanded, and the configuration **Timer_WOI_debug_flash_jlink** is selected. The **Debugger** tab is active, showing the **J-Link GDB Server Setup** and **GDB Client Setup** sections. The **J-Link GDB Server Setup** section includes fields for **Executable**, **Device name** (S32K144), **Endianness** (Little), **Connection** (USB), **Interface** (JTAG), **Initial speed** (Fixed, 1000 kHz), **GDB port** (2331), **SWO port** (2332), **Telnet port** (2333), **Log file**, and **Other options**. The **GDB Client Setup** section includes fields for **Executable**, **Other options**, and **Commands**. The **Remote Target** section is empty. The **Debug** button is highlighted at the bottom right of the dialog.

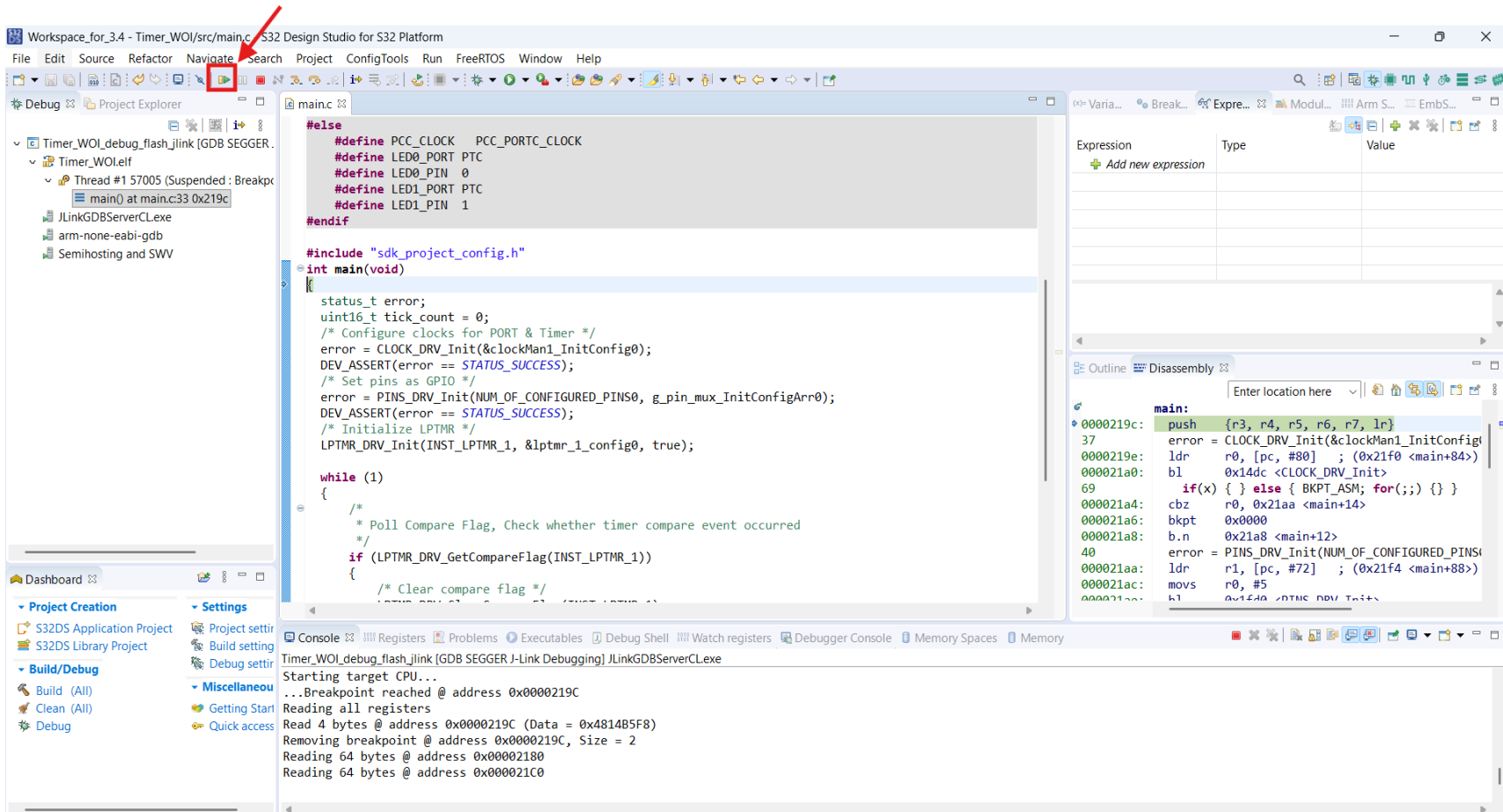
- Click on Accept



- Click on Switch, changes to Debug perspective



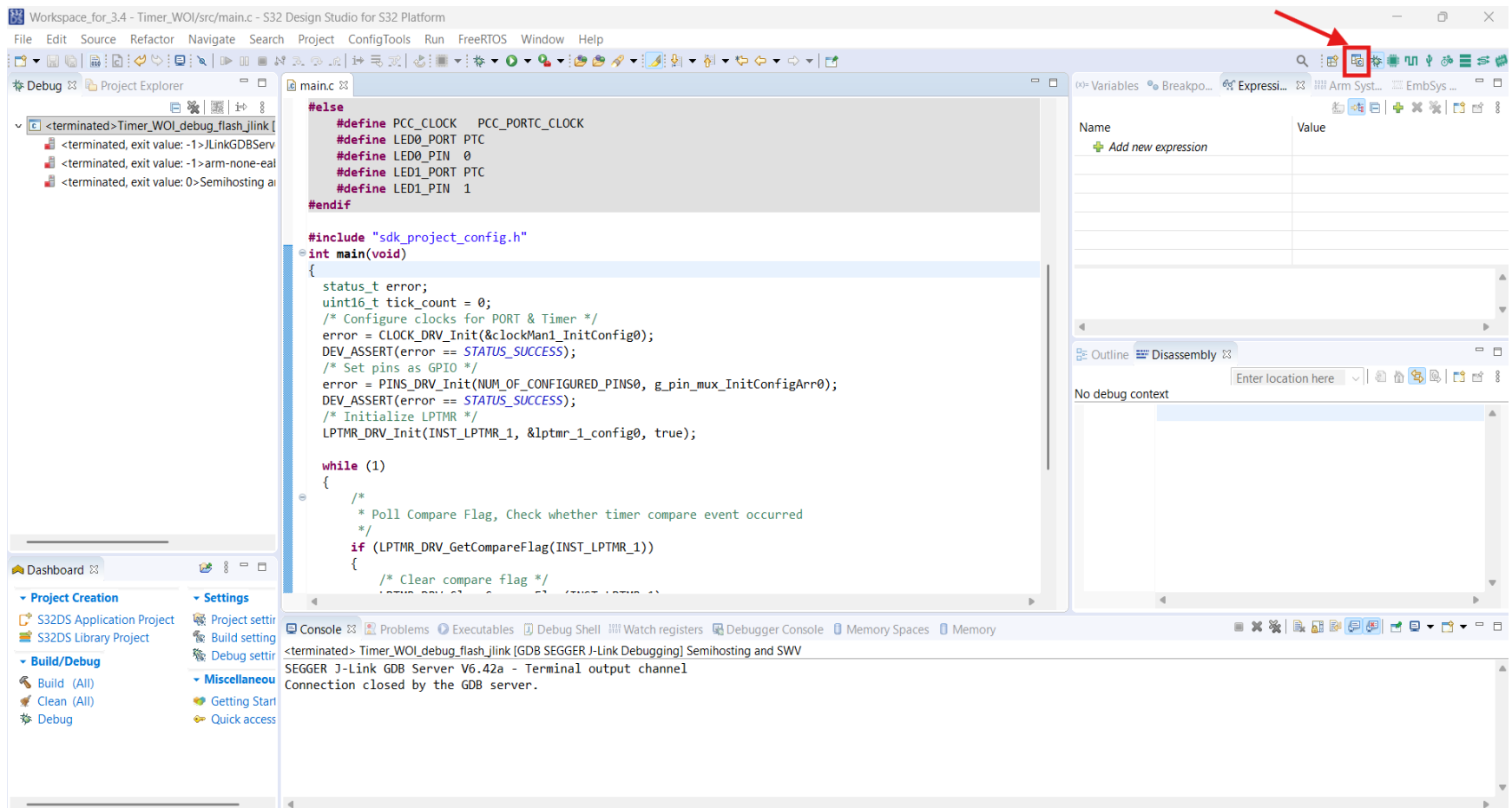
- To run code on board, click on Resume and see output on board



- Click on Terminate to stop debugging

The screenshot shows the S32 Design Studio IDE interface. The top toolbar contains various icons, with the 'Terminate' icon (a red square with a white 'X') highlighted by a red arrow. The main editor displays the source code for `main.c`, which includes definitions for clock and pin configurations, and a `main` function that initializes the system and enters a loop. The right-hand side of the IDE features a 'Disassembly' window showing the assembly code for the `main` function, with addresses ranging from `0000219c` to `000021ac`. The bottom status bar indicates the current state of the debugger, showing 'Thread #1 57005 (Running : User Request)' and 'JLinkGDBServerCL.exe'.

- Click on C/C++ to change perspective



Workspace_for_3.4 - Timer_WOJ/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Run Window Help

Project Explorer

- hello_world_s32k144
- Timer_WOJ: Debug_FLASH
 - Binaries
 - Includes
 - Project_Settings
 - SDK
 - board
 - src
 - main.c
 - Debug_FLASH
 - Debug_Configurations
 - doc
 - Doxygen
 - description.txt
 - Timer_WOJ.mex

Dashboard

- Project Creation
 - S32DS Application Project
 - S32DS Library Project
- Miscellaneous
 - Getting Started
 - Quick
- Build/Debug
 - Build (All)
 - Clean (All)
 - Debug
- Settings
 - Project settings
 - Build settings
 - Debug settings

main.c

```

#else
#define PCC_CLOCK    PCC_PORTC_CLOCK
#define LED0_PORT    PTC
#define LED0_PIN     0
#define LED1_PORT    PTC
#define LED1_PIN     1
#endif

#include "sdk_project_config.h"
int main(void)
{
    status_t error;
    uint16_t tick_count = 0;
    /* Configure clocks for PORT & Timer */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Initialize LPTMR */
    LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);

    while (1)
    {
        /*
         * Poll Compare Flag, Check whether timer compare event occurred
         */
        if (LPTMR_DRV_GetCompareFlag(INST_LPTMR_1))
        {
            /* Clear compare flag */
            LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);
            /* Increment software counter */
            tick_count++;
        }
    }
}

```

Outline

- # EVB
- # PCC_CLOCK
- # LED0_PORT
- # LED0_PIN
- # LED1_PORT
- # LED1_PIN
- # PCC_CLOCK
- # LED0_PORT
- # LED0_PIN
- # LED1_PORT
- # LED1_PIN
- sdk_project_config.h
- main(void): int

Problems Tasks Console Properties

<terminated> Timer_WOJ_debug_flash_jlink [GDB SEGGER J-Link Debugging] Semihosting and SWW
SEGGER J-Link GDB Server V6.42a - Terminal output channel
Connection closed by the GDB server.

Kind Courtesy of reference books/papers/Links:

1. S32K1xx Series Reference Manual
2. S32SDK User Manual - S32K1xx RTM 4.0.3
3. <https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-s32-platform:S32DS-S32PLATFORM>

Thank you